

Operation-Aware Biomedical Text Simplification

Preliminary Report — Operation Classification and Baselines

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1 Operation Classification

The central contribution of our system is a *diagnostic* step: before simplifying a span of biomedical text, we predict *which* simplification operation it requires. We adopt the three-operation taxonomy of Heineman et al. [1]: **Substitution** (replacing a jargon term with a plain-language equivalent), **Explanation** (adding words to unpack a concept), and **Generalization** (compressing or removing detail). While prior PLABA systems apply a single undifferentiated simplification model, we argue that routing each span to the appropriate operation is what preserves safety-critical information.

Pseudo-labeling. Span-level operation labels are not available in PLABA, so we derive them by weak supervision from the source→target sentence pairs. The *type* of edit a human writer performed leaves a measurable fingerprint on the text: an Explanation lengthens the sentence, a Generalization shortens it, and a Substitution changes many characters while keeping the length roughly constant. Concretely, let $r = |t|/|s|$ be the target/source word-length ratio. We label a pair Explanation if $r \geq 1.15$, Generalization if $r \leq 0.85$, and otherwise Substitution, using normalized character-level edit distance to confirm that similar-length pairs were genuinely rewritten rather than copied. Edit distance is the most direct measurable signal of the kind of change made, and requires no gold annotation — standard weakly-supervised practice.

Data and leakage control. The PLABA release (`data.json`) is the full corpus and contains the abstracts held out in the official validation and test splits. To avoid train/test leakage, we exclude every validation and test PMID from the training pairs at the source level. This yields **6,307** pseudo-labeled training sentence pairs, distributed across the three operations as shown in Table 1. We evaluate on **1,431** sentence pairs built from the held-out validation PMIDs and pseudo-labeled identically.

Operation	Train pairs	Share
Substitution	2,723	43.2%
Explanation	2,167	34.4%
Generalization	1,417	22.5%
Total	6,307	100%

Table 1: Pseudo-labeled operation distribution on the leakage-free training set.

Three-model progression. We plan a controlled progression of classifiers, each isolating one variable: (i) **TF-IDF + Logistic Regression** tests whether surface lexical features alone predict the operation; (ii) **BERT-base** [3] adds general contextual pretraining; and (iii) **BioBERT** [4] adds biomedical-domain pretraining on top. Reporting all three localizes exactly where any gains originate. This preliminary report covers the TF-IDF baseline; the contextual models follow in the full paper.

Preliminary results. The classifier receives only the *source* sentence, since in the full pipeline the operation must be chosen before the simplified text is generated. Using (1,2)-gram TF-IDF features (`max_features` = 50,000) and Logistic Regression with balanced class weights, the model attains **0.431** accuracy and **0.424** macro-F1 on the validation set (Table 2). Although accuracy is close to the majority-class rate (0.427), the macro-F1 is far above the corresponding majority baseline (≈ 0.20), confirming that the model makes balanced predictions across all three classes rather than collapsing to the majority. The confusion matrix (Table 3) shows the strongest confusion between Substitution and Explanation, consistent with the fact that both often preserve sentence length. These results establish a weak-but-nontrivial lower bound and motivate the contextual models, which can capture semantic and structural cues unavailable to bag-of-words features.

	Precision	Recall	F1
Substitution	0.471	0.481	0.476
Explanation	0.427	0.351	0.385
Generalization	0.376	0.449	0.410
Macro avg	0.425	0.427	0.424
Accuracy	0.431 (n = 1,431)		

Table 2: TF-IDF + Logistic Regression, per-class results on the validation set.

True \ Pred	Sub	Exp	Gen
Substitution	294	150	167
Explanation	203	163	98
Generalization	127	69	160

Table 3: Confusion matrix (rows = true, columns = predicted).

2 Baselines

We frame our baselines as a comparison ladder anchored on *warning preservation* — the fraction of source warning phrases that survive simplification. This metric targets safety-critical information loss that n-gram metrics such as ROUGE ignore.

Baseline 1 (no simplification) leaves the text unchanged and thus trivially preserves all warnings (1.000); it is the safety ceiling against which no system should be expected to improve. **Baseline 2 (rule-based CHV substitution)** performs Consumer Health Vocabulary term replacement and sentence splitting, preserving 99.4% of warnings — auditable and low-risk, but limited to vocabulary-level edits. **Baseline 3 (direct LLM prompting)** is the deliberately naive

Baseline	Warning preservation
No simplification (ceiling)	1.000
Rule-based CHV substitution	0.994
Direct LLM prompting (FLAN-T5-Base)	0.950
Operation-aware system (ours)	TBD

Table 4: Warning preservation across baselines. Direct LLM prompting loses the most safety-critical information.

approach our system argues against: we prompt an instruction-tuned open-source LLM (FLAN-T5-Base) to “simplify this biomedical sentence for a patient,” with no operation guidance and no safety constraint. We run it locally to keep the baseline fully reproducible (no paid API).

On a 30-sentence sample restricted to warning-bearing sentences, direct LLM prompting preserved only **95.0%** of source warnings — measurably below both the rule-based baseline (99.4%) and the do-nothing ceiling (100%). This confirms our central premise: unguided LLM simplification sacrifices safety-critical information that conservative methods retain, motivating the operation-aware routing our full system introduces. The three baselines together bracket the safety–fluency trade-off our system must navigate.

Literature Review

SALSA edit taxonomy [1] defines the fine-grained edit operations that underpin our three-way operation taxonomy, directly grounding our Substitution/Explanation/Generalization labels. The **PLABA dataset** [2] provides the sentence-aligned source–adaptation pairs we pseudo-label. **BERT** [3] and **BioBERT** [4] motivate the contextual and biomedical-domain classifiers in our model progression. The **TSAR shared task** [5] documents the field’s shift toward LLM-based simplification and the accompanying safety gaps that motivate our diagnostic, safety-first framing.

References

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